

Foundation (Jalapeno)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x + y = 2$. What is x ?
(A) 0
(B) 1
(C) 2
(D) -1
(E) -2
- Q2.** A company's profit is modeled by $P = x^2 - 4x + 5$ and cost by $C = x^2 + 2x + 3$. What is the break-even point?
(A) (-1, 4)
(B) (-2, 0)
(C) (1, 4)
(D) (2, 9)
(E) (0, 5)
- Q3.** Which system has no real solutions: $x^2 + y^2 = 1$, $x^2 - y^2 = 4$?
(A) No solution
(B) Infinite solutions
(C) One solution
(D) Two solutions
(E) More than two
- Q4.** If $x^2 - 2y = 3$ and $y = x^2 - 5$, what is x ?
(A) -1
(B) 1
(C) -2
(D) 2
(E) No solution
- Q5.** What is the point of intersection between $y = x^2$ and $y = x + 1$?
(A) (0, 0)
(B) (1, 2)
(C) (-1, 0)
(D) (-1, 2)
(E) (0, 1)
- Q6.** Solve the system of equations: $y = x^2 - 3$ and $y = x - 1$. What is the x -coordinate of the point of intersection?
(A) -1
(B) 1
(C) 2
(D) -2
(E) No real solution
- Q7.** A stock's price is modeled by $P = 2x^2 + 5x + 1$ and the number of shares sold by $S = x^2 + x + 2$. What is the revenue when $x = 1$?
(A) 16
(B) 24
(C) 32
(D) 48
(E) 64
- Q8.** An investor has two investment options with returns given by $R = x^2 + 2x + 3$ and $R = x^2 - x + 1$. When do they have the same return?
(A) $x = 0$
(B) $x = 1$
(C) $x = -1$
(D) $x = -2$
(E) $x = -3$

Open Ended Questions (FRQ)

- Q9.** A company has two factories. The production cost of the first factory is $C_1 = x^2 + 5x + 10$ and the production cost of the second factory is $C_2 = x^2 - 2x + 8$. Find the point of intersection of the two cost functions if x represents the number of units produced. Show your work and explain your answer.
- Q10.** If a stock's price is modeled by $P = x^2 + 3x + 2$ and the number of shares sold is modeled by $S = x^2 - 2x + 1$, find the revenue as a function of x and graph the system of equations. What does the point of intersection represent in this context? Explain your answer.